

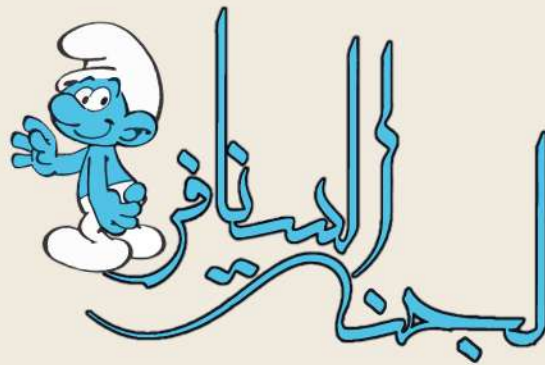
خدمتكم طريق خضناه لرضى الله

2021

أسئلة سنوات

ميد

إحصاء وإحتمالات



سنافر البولي تكنك f

بسم الله الرحمن الرحيم

نقدم لكم نحن أسرة فريق
(لجنة السنافر)

مجموعة أسئلة اختبارات إلكترونية تم تجميعها خلال الفصول الماضية
سائلين المولى أن يوفقنا وإياكم لكل خير

تنويه

يوجد بعض الأسئلة عليها إجابات قد تحتل الصواب وقد تحتل الخطأ
فإن أصبنا فما هو إلا توفيق من الله
وإن أخطأنا فمن أنفسنا

#خدمتكم_طريق_خضناه_لرضى_الله

#الإتجاه_الاسلامي

#بسواعدنا_نبنيها

#لجنة_السنافر

#هي_الله

For random variables X and Y , we have
 $E(X) = 1$, $E(Y) = 2$ and
 $E(XY) = 1$; then $Cov(X, Y) = 1$.

Select one

☐ True

☒ False

For some random variable X , assume that $E(X) = 1$, $E(X^2) = 1$ and $E(X + X^2 + X^3) = 6$; then $E(X^3) =$

Select one:

☐ a. 3

☒ b. 4

☐ c. 2

☐ d. 1



For a random variable X with mean $E(X) = \mu$ and variance σ^2 , we have $E(X - \mu) = 0$.

Substituting

$\mu = 0$

$$X\sigma^2$$

$$X^2\sigma^2$$

$$0$$

$$d$$

$$\sigma^2$$



Let X and Y be some random variables ; then $E(\overline{X}Y) = \overline{X}\overline{Y}$

Select one:

☐ True

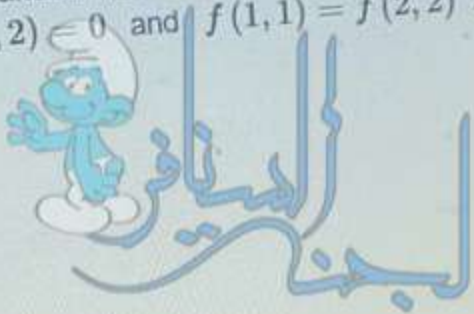
☒ False



Let X and Y be some random variables with range $R_{XY} = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.
If $f(2, 1) = f(1, 2) = 0$ and $f(1, 1) = f(2, 2)$. Then $f(1, 1) =$

Select one:

- ☐ a. 0.2
- ☒ b. 0.5
- ☐ c. 0.1
- ☐ d. 0.3



Let X be a continuous random variable with pdf $f(X) = e^{-X}$ when $X > b - 1$ and zero otherwise. The constant b equals

Select one:

☒ a. 1

☐ b. 3

☐ c. 2

☐ d. 0



Randomly we draw 2 balls from a box that contains 5 red balls and 2 black ball. The probability of choosing 2 red balls is

$$\frac{\binom{5}{2} \binom{2}{0}}{\binom{7}{2}}$$

Select one:



True



False



الاحتمال

For some positive constants a and b , let X and Y be some random variables with joint pdf

$$f(x, y) = \begin{cases} 1 & , 0 \leq x \leq a, 0 \leq y \leq b \\ 0 & , \text{Otherwise} \end{cases}$$

The marginal pdf with respect to X is

$$f_X(x) = a; \text{ if } 0 \leq x \leq a$$

Select one:

☐ True

☒ False

For two **disjoint** events A and B with $P_r(A) = 0.3$ and $P_r(B) = 0.7$; we have $P_r(A - B) =$

Select one:

☒ a. 0.4

☐ b. 0.2

☐ c. 0.3

☐ d. 0.1



For any event A - we have $P(A|\bar{A})=0$

Select one:

☒ True

☐ False



الاجابة
لـ بـ جـ نـ

Let X and Y be some random variables with range $R_{XY} = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$. If $f(1, 1) = f(2, 2) = 0$ and $f(2, 1) = f(1, 2) = f(2, 1)$. Then



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a. 0.5 ☒

b. 0.2 ☐

c. 0.3 ☐

d. 0.1 ☐

For some independent random variables, X and Y assume that $E(XY) = E(X) = 0$. Then $E(Y) = 1$.

Select one

☐ True

☒ False



For some **NONEMPTY** independent events A and B we have $P_r(A) = P_r(B) = P_r(A \cap B)$, then

Select one:

☒ a. Event A is impossible and event B is certain

☐ b. Both A and B are impossible events

☐ c. Both A and B are certain events

☐ d. Event A is certain and event B is impossible



10 April - MID EXAM

For some independent random variables X and Y assume that $E(XY) = E(X) > 0$ then $P(Y) = 1$

Select one:

☐ True

☒ False



For random variable X with mean $E(X) = \bar{X}$ and variance σ^2 , we have $Var(-\bar{X} - X) =$

Select one:

☐ a.

0

☐ b.

$\bar{X}^2 \sigma^2$

☐ c.

σ^2

☐ d.

$\bar{X} \sigma^2$



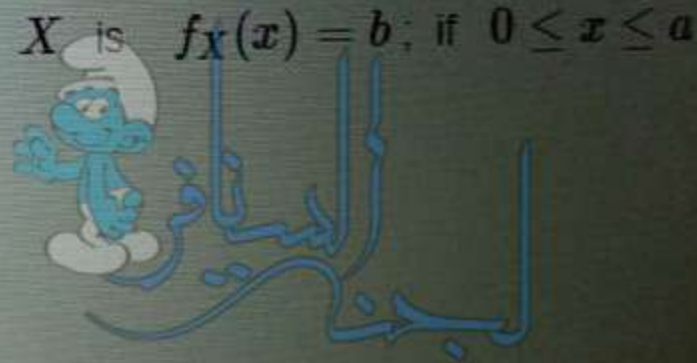
For some positive constants a and b , let X and Y be some random variables with joint pdf

$f(x, y) = \begin{cases} 1 & , 0 \leq x \leq a, 0 \leq y \leq b \\ 0 & , \text{Otherwise} \end{cases}$. The marginal pdf with respect to X is $f_X(x) = b$; if $0 \leq x \leq a$

Select one:

☒ True

☐ False



Let X and Y be some random variables with range $R_{XY} = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.
If $f(2, 1) = f(1, 2) = 0$ and $f(1, 1) = f(2, 2)$. Then $f(1, 1) =$

Select one:

☐ a. 0.2

☒ b. 0.5

☐ c. 0.1

☐ d. 0.3



For random variables X and Y assume that $\text{Corr}(3X, -Y) = \text{Corr}(X, Y)$. Then X and Y are independent.

Select one:

☐ True

☒ False



$E(X^3)$ = For some random variable X , assume that $E(X) = 1$, $E(X^2) = 1$ and $E(X + X^2 + X^3) = 3$; then



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a. 1 ☐

b. 2 ☐

c. 4 ☒

d. 3 ☐

For random variables X and Y , we have
 $E(X) = 1$, $E(Y) = 2$ and
 $E(XY) = 4$ then $Cov(X, Y) = 1$.

Select one:

☐ True

☒ False



For random variables X and Y assume that $\text{Corr}(3X, -Y) = \text{Corr}(X, Y)$. Then X and Y are independent.

Select one

☒ True

☐ False

For some positive constants a and b , let X and Y be some random variables with joint pdf

$$f(x, y) = \begin{cases} 1 & , 0 \leq x \leq a, 0 \leq y \leq b \\ 0 & , \text{Otherwise} \end{cases}$$

The marginal pdf with respect to X is $f_X(x) = b$; if $0 \leq x \leq a$

Select one:

☒ True

☐ False

$P_r(A \cup B) = 2\lambda$ For some disjoint events A and B , assume that $P_r(A) = P_r(B) = \lambda$, we have



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☐ صحيح

☒ خطأ

For events A and B , assume that B is the impossible event. If $P_r(A - B) = 0.1$; then $P_r(A | \bar{B}) =$

Select one:

- ☒ a. 0.1
- ☐ b. 0.4
- ☐ c. 0.3
- ☐ d. 0.2

For some random variable X , assume that
 $E(X) = 1$, $E(X^2) = 1$ and
 $E(X + X^2 + X^3) = 4$; then
 $E(X^3) =$

Select one:

☒ a. 2

☐ b. 3

☐ c. 1

☐ d. 4

Two continuous and independent random variables X and Y with pdfs

$$f_X(x) = 2 \quad \text{when } 0 < x < \frac{1}{2} \quad \text{and zero otherwise}$$

$$f_Y(y) = \frac{1}{2}e^{-2y} \quad \text{when } y > 0 \quad \text{and zero otherwise}$$

Then the joint pdf is $f(x, y) = 2e^{-2y}$ when $0 < x < \frac{1}{2}$ and $y > 0$ and zero otherwise

Select one:

☒ True

☐ False

For two **disjoint** events A and B with $P_r(A) = 0.2$ and $P_r(B) = 0.7$; we have $P_r(A - B) =$

Select one:

☒ a. 0.4

☐ b. 0.1

☐ c. 0.3

☐ d. 0.2